

Reading - Ch 6, CASI - Efron & Hastie

- Empirical Bays started developing in 1940's

KEY QUESTIONS TO ANSWER:

- ① What is the core problem being solved?
 - unknown prior distribution, but can now use observed data to assume prior using empirical dist
- ② What is Robbins' Formula? What does it do?
 - specifically uses the empirical dist. of your observed data to estimate what the prior dist is.
- ③ What's an example from the text?
 - Missing Species
 - Butterfly or Shakespeare
 - Medical → Cancer surgeries
- ④ What confuses me?
 - some of the formulas

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- modern technology really provided the push for understanding of empirical Bayes

6.1 Robbins' Formula -

- Companies/People always curious what will happen next based on the data from now.

Robbins' Formula $\rightarrow E\{\theta|x\} = (x+1)f(x+1)/f(x)$

$\hookrightarrow$ allows for an estimate of the expectation without needing prior distributions of the data
marginal density $f(x) \rightarrow$ proportion of total counts in

$$\hat{f}(x) = \frac{y_x}{N} \quad \text{category } x.$$

$\Rightarrow$ yields an empirical version of Robbins' Formula

$$\hat{E}\{\theta|x\} = (x+1)\hat{f}(x+1)/\hat{f}(x) = (x+1)y_{x+1}/y_x$$

* destabilized
by small
counts

so the final expression doesn't need N , which
is total count of $\sum y_x$.

Q: So Robbins' Formula is just using the count data you
have to estimate a prior expectation?

A1: - uses the ^{empirical} distribution of your observed counts to
estimate what the prior distribution should be.

Then, uses that prior distribution & Bayes' to shrink
each estimate toward the expected value

Empirical Bayes:

- "the Bayesian formula for a single observation
is estimated empirically from a collection of
similar cases."

critical
point

- "Large data sets of parallel situations
carry with them their own Bayesian information"

* When destabilized by small counts, now assume
a parametric prior density distribution for
more dependable results.

- use parametric prior density $\rightarrow$ then estimates for the parameters found through maximum likelihood fitting counts $\&$ yielding a parametrically estimated marginal density $\hat{f}(x) = f_{\hat{\theta}, \hat{\alpha}}(x)$ or $\hat{f}_k = 1/N f_{\hat{\theta}, \hat{\alpha}}(x)$
- * works better as x gets larger fixing the destabilizing issue.

6.2 Missing-Species Problem

Butterfly Data problem # of y species seen/trapped
 x number of times

Question Corbet asked $\rightarrow$ about future expectation.

How many new species if trapped for additional year?

* The unknown of the # of unseen species ($x=0$)

S species (seen or unseen), $X_k = \#$ of times species k is seen in one time unit

independent $\rightarrow X_k \sim \text{Pois}(\theta_k)$ for $k=1, \dots, S$ further trapping
 $Y_k = \# \{X_k = x\}$ for $x=1, 2, \dots, 24$, $t = \text{time units}$

$X_k(t) \sim \text{Pois}(\theta_k t)$ $t = 1/2$ (b/c 1 more yr out of 2 past)

* $X_k(t)$ independent of X_k $\rightarrow$ any one species is trapped independently over time at a rate proportional to its parameter θ_k .

looking for when $X_k=0$ and $X_k(t) > 0 \rightarrow e^{-\theta_k}(1 - e^{-\theta_k t})$

take expectation $\&$ use empirical density $\frac{1}{S}$ on each θ_k values

* don't know θ values but can estimate like in Robbins' with the y observed values.

- calculate standard error of $\hat{E}(U)$ if willing to assume that x_k 's are mutually independent.
- * but process somewhat unstable because using non-parametric y_k 's to estimate e_k
- parametric form used $g(\theta)$ assumed to be gamma then take $\hat{e} = y_i$ using MLE gives gamma parameters
- * Similar results but parametric more "reasonable" which not same as dependable.
- watch out for ~~educated~~ educated guessing too far into future.

Also - Shakespeare (884,647 words in canon)
31,534 distinct words

→ SO, what words did Shakespeare know but not use??
↳ these are "missing species"

- suppose new manuscripts found w/ 884647 $\cdot t$ words
($t=1$ is new canon just as large)

⇒ Missing Mass (another type of problem)

$$M = \frac{\sum_{\text{unseen}} \theta_k}{\sum_{\text{all}} \theta_k} = \frac{e_1}{N} \quad \hat{M} = \frac{y_1}{N}$$

6.3 Medical Example

(+) = malignant (cancer)

$N = 844$ surgeries, $n = \#$ nodes removed, $x = \#$ nodes (+)

$$p_k = \frac{x_k}{n_k} \quad k=1,2,3,\dots,N$$

- say patients have some true probability of a node being positive, & node results are independent of one another so

$$X_k \sim \text{Bin}(n_k, \theta_k)$$

so $p_k = \frac{X_k}{n_k}$ has $p_k \sim (\theta_k, \frac{\theta_k(1-\theta_k)}{n_k})$

θ_k more accurately estimated when n is large.

Bayesian analysis $\rightarrow$ need assumption of prior density function $g(\theta)$ for θ_k

$\rightarrow$ since we don't know $g(\theta)$, so empirical Bayes.

- Based on histogram, prior could be 4th degree poly.

- Then marginal probability of X_k for patient k is binomial so take MLE.

Posterior dist of θ_k estimated according to Bayes' rule
 $\rightarrow$ use to estimate patients.

6.4 Indirect Evidence

- highlighted on text.